

1. Problem Statement & Motivation

Real-World Business Problem

Consumers frequently face challenges when making purchasing decisions due to the large volume of inconsistent and fragmented product information available across online retailers. Product specifications are often presented in different formats, reviews may be biased or contradictory, and pricing varies across platforms and over time. This creates inefficiencies in the decision-making process, leading consumers to spend excessive time researching products or make purchases that do not fully align with their needs and constraints. From a business perspective, poor purchase decisions can reduce customer satisfaction, increase product return rates, and weaken long-term trust in digital marketplaces.

Target Users

The primary target users for this system are students, budget-conscious consumers, and everyday online shoppers who need to compare multiple products efficiently. These users typically value clear, transparent recommendations and do not want to manually analyze technical specifications, reviews, and price differences across several websites. The system is particularly useful for consumers purchasing technology and consumer electronics, where tradeoffs between cost, performance, and features are common.

Why an Agentic AI Solution Is Appropriate

An agentic AI solution is appropriate because purchase comparison is inherently a goal-oriented and multi-step reasoning task that cannot be effectively handled by static comparison tools or

predefined workflows. The system must interpret user-defined constraints, gather and normalize product data, evaluate tradeoffs, and generate ranked recommendations that adapt to changing user preferences. Unlike a human-defined workflow or a one-time chatbot response, an agentic system can autonomously plan actions, coordinate subtasks, and critique its own outputs, making it more flexible, scalable, and aligned with real-world decision-making processes.

2. Use Cases & Scenarios

Scenarios

In representative user scenarios, a user wants to compare multiple products under specific constraints such as budget, performance requirements, and intended use. For instance, a student may want to select a laptop within a fixed price range for coursework and programming, a commuter may want to choose headphones optimized for noise cancellation and call quality, or a user may want to compare monitors that balance productivity and gaming performance. In each scenario, the agentic system interprets user constraints, retrieves or processes product information, extracts and normalizes specifications and pricing data, evaluates options against user-defined priorities, identifies tradeoffs and potential risks, and generates a ranked comparison with clear explanations and alternative recommendations when constraints conflict.

3. Initial System Design

Agent Roles

The system is designed around a multi-agent architecture coordinated by a central planning component. A coordinator or planner agent manages the overall workflow by validating user inputs, sequencing tasks, and integrating outputs from other agents. A specification analyzer

agent is responsible for extracting and normalizing product specifications from user-provided descriptions or external sources. A price tracking agent retrieves and compares pricing information to ensure recommendations remain within budget constraints. Finally, a value critic agent evaluates the overall value-for-money of each product, challenges initial rankings, and flags hidden tradeoffs or risks that may not be immediately apparent.

Expected Tools and Data Sources

The system is expected to use web retrieval or search tools to access product information when links are provided, along with local or static datasets such as CSV or JSON files for controlled testing and evaluation. Users may also upload product descriptions or specifications directly. A lightweight front-end interface, such as a Streamlit or web-based application, will allow users to input products and constraints and view ranked recommendations in an accessible format.

4. Feasibility & Risks

Potential Challenges

Several challenges may arise during implementation, including incomplete or inconsistent product specifications across sources, the risk of hallucinated information when data is missing, and potential failures or limitations in external tools used for data retrieval. Additionally, evaluating product value can be subjective, as different users may prioritize features differently.

Constraints

The system must operate within practical constraints such as external tool rate limits, limited computation resources, and the course's time frame. Privacy considerations also apply, as the system will not store or process sensitive personal information beyond user-defined preferences.

Project Timeline

During the first two weeks, the project will focus on proposal finalization and system design.

The third week will be dedicated to building the front-end prototype for Milestone I. In the fourth week, the multi-agent workflow will be implemented to satisfy Milestone II requirements.

The fifth week will focus on designing the evaluation plan and conducting pilot testing for Milestone III. The final week will be used to complete the full system implementation, prepare the written report, and record the final presentation.